Impacts of H3K9me2 in Schizophrenia

Isabella Castellano, Xander Krohannon, MS.

Introduction

- Schizophrenia (SCZ) is a chronic psychiatric disorder driven in part by epigenetic dysregulation in the brain
- H3K9me2, a repressive histone mark, is understudied yet suspected to alter gene expression in SCZ
- We investigated whether H3K9me2 modifications correlate with differential gene expression in the PFC (Brodmann areas 9 & 10) of SCZ patients

Data

- ChIP-seq (H3K9me2): 15 SCZ, 15 controls
  - PRJNA891668, Frontiers in Psychiatry (2022)
- RNA-seq: 19 SCZ, 18 controls
  - PRJNA695206, Schizophrenia Bulletin Open (2021)

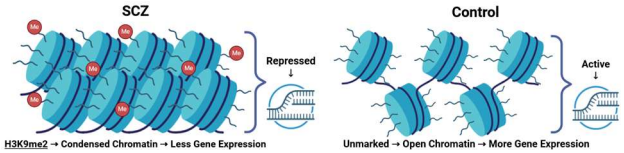
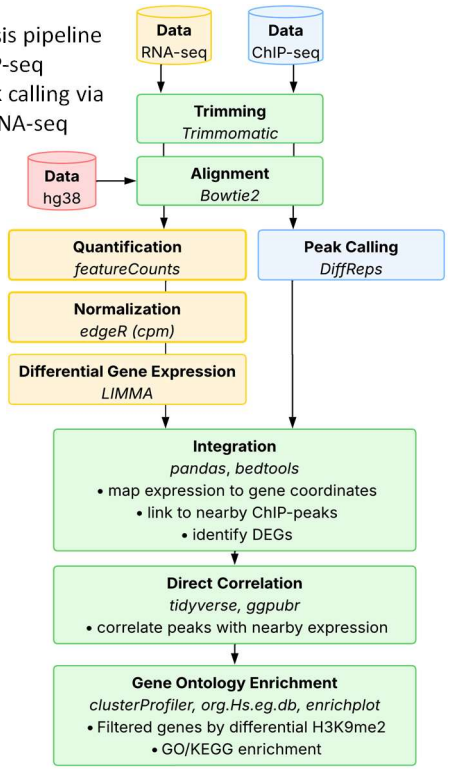


Figure 1: Differences in effects of histone modifications

Figure 2: Analysis pipeline integrating ChIP-seq (H3K9me2 peak calling via DiffReps) and RNA-seq (differential expression via LIMMA) to identify genes with epigenetic and transcriptional changes in schizophrenia



Scan for GitHub

Scan for references

Effects of H3K9me2 on Gene Expression in Schizophrenia

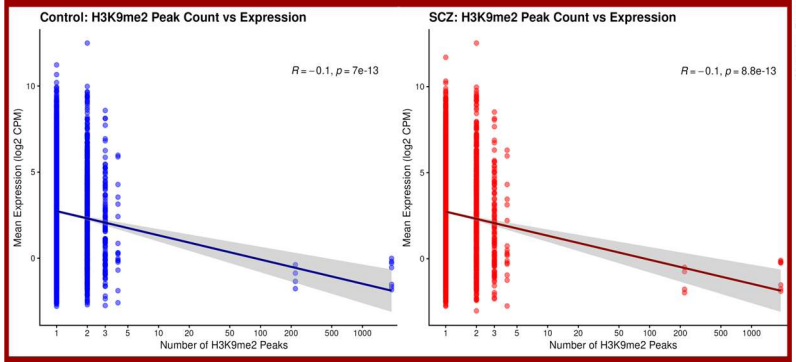


Figure 3: Scatter plot of H3K9me2 peak count versus mean gene expression for control cases (left) and a schizophrenia cases (right)

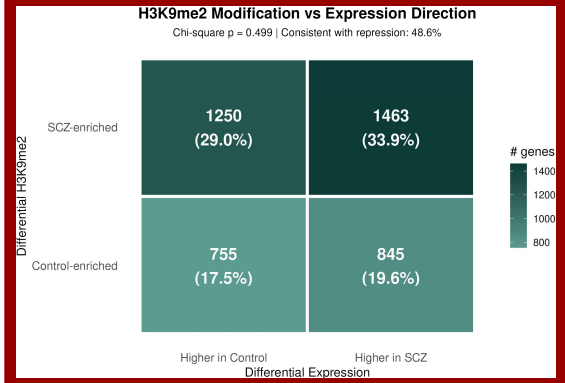


Figure 4. Confusion matrix comparing direction of differential H3K9me2 modification (SCZ-enriched vs Control-enriched) against direction of differential expression (higher in SCZ vs higher in Control)

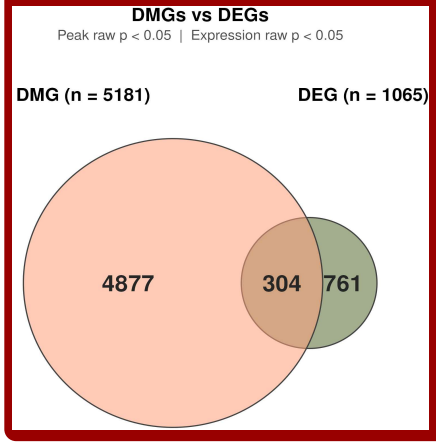


Figure 5: Venn diagram of differentially modified genes (DMGs; H3K9me2 peaks, raw $p < 0.05$) and differentially expressed genes (DEGs; raw $p < 0.05$) in SCZ vs control. 304 genes show both differential H3K9me2 modification and differential expression, representing candidate genes for epigenetic regulation in schizophrenia

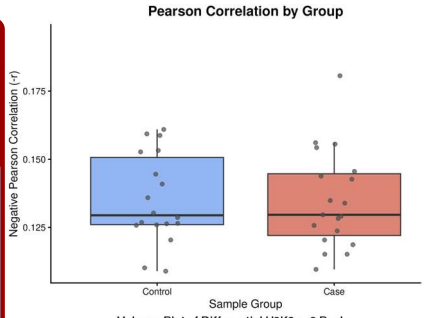


Figure 6: Box plot showing negative Pearson correlation values between H3K9me2 signal and expression of each peak's nearest gene grouped by sample type

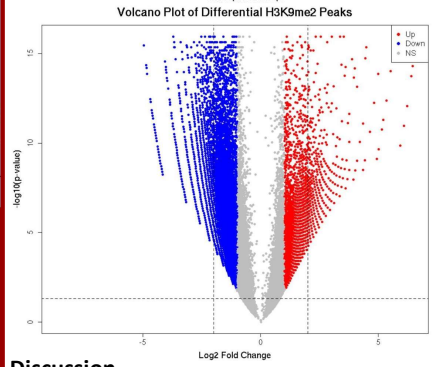


Figure 7: Volcano plot of differential H3K9me2 peaks between SCZ and control samples. Peaks with a significant increase of signal in SCZ are shown in red (Up), decreased in blue (Down), and non-significant in gray (NS)

Discussion

- H3K9me2 was negatively correlated with gene expression, consistent with its role as a repressive histone mark
- SCZ samples showed significantly more H3K9me2 markers per gene than controls, suggesting elevated repression
- However, direction of H3K9me2 change at individual genes did not predict direction of expression change
- GO of overlapping DMGs/DEGs yielded no significant results

Conclusion

- H3K9me2 acts as a global repressive mark in both cohorts
- SCZ brains show a global increase in H3K9me2 markers, but this does not translate to predictable per-gene expression changes
- This suggests H3K9me2 operates at the domain level rather than as a precise per-gene switch

Future Directions

- Create a co-modification network to identify relationships between genes that are modified simultaneously
- Perform GSEA would rank genes by modifications

Limitations

- ChIP-seq and RNA-seq came from different cohorts & brain regions
- H3K9me2 acts over broad heterochromatin domains, not single genes
- Many other factors (other marks, TFs, DNA methylation) regulate expression